

DEEPAK LEKHAK

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Technical Skills

Languages: Python, Java, C, HTML/CSS, JavaScript, SQL, Bash

Developer Tools: VS Code, PowerBI, Looker Studio, IntelliJ, Docker

Technologies/Frameworks: D3.js, Pandas, Matplotlib, PostgreSQL, MongoDB, MySQL, TensorFlow, NumPy

Certifications: IBM Data Science Professional Certificate, Google IT Support Professional Certificate

Experience

University of Waterloo

May 2023 – Aug. 2023

Data Analyst, Business Intelligence

Waterloo, ON

- Built interactive **Power BI** dashboards and **KPI reports**, enabling **trend analysis** and data-driven decisions.
- Developed predictive models using **hypothesis testing** and **scenario analysis** to improve AI system design and user experience.
- Evaluated data storage, encryption, and cloud solutions; implemented **Azure SQL**, **dynamic data masking**, and **Key Vault** for scalability and compliance.

Projects

Employee Churn Prediction 📊 | *Python, PyCaret, BigQuery, Looker Studio*

Aug. 2025 – Sept. 2025

- Achieved **98.8% accuracy** and **0.99 AUC** with a **Random Forest classifier**, outperforming 10+ baselines (Logistic Regression, SVM) via cross-validation.
- Ranked **job satisfaction** as top churn driver (**31.7% importance**) through feature engineering and model interpretation, guiding HR interventions.
- Processed **15K+ employee records** from **BigQuery** with pipelines for imputation, encoding, and balancing; published predictions and feature tables back to BigQuery for real-time reporting.
- Built **Looker Studio dashboards** showing churn by **department**, **salary**, **tenure**, providing executives with insights across 10+ departments.

HR Analytics Dashboard 📊 | *Power BI, Data Visualization, HR Analytics*

Jul. 2025 – Aug. 2025

- Built an **interactive Power BI dashboard** analyzing **5,124 absenteeism hours** across **740 cases**, enabling HR to monitor trends by **employee category**, **weekday**, and **season**.
- Identified **Category 3 employees** as highest-risk with **765 hours (14.9%)** of total absenteeism, guiding workload balancing and retention strategies.
- Analyzed top causes of absenteeism (**149 medical**, **112 dental**, **69 physiotherapy**) to uncover health-related drivers, supporting wellness initiatives.
- Visualized absenteeism peaks by **weekday (1,489 hours)** and seasonal patterns (Fall, Winter), enabling better resource planning and policy adjustments.

Pundit Statbook 📊 | *Next.js, TypeScript, D3.js, React*

Mar. 2025 - Apr. 2025

- Built a responsive football analytics dashboard with role-based pages for Defenders, Midfielders, Forwards, and Goalkeepers.
- Implemented interactive radar charts with **D3.js** for player and system metrics, including animations, tooltips, and dynamic resizing.
- Developed modular React components (**RadarChart**, **SystemRadarChart**, grids) and leveraged Next.js SSR for fast performance.

FaceNet CNN Implementation 📊 | *Python, TensorFlow, Keras, FaceNet*

Jan. 2024 - Apr. 2024

- Implemented **FaceNet architecture** in **TensorFlow/Keras**, building custom convolutional and embedding layers from scratch.
- Designed a training pipeline with **triplet loss**, preprocessing, and **mini-batch sampling**, improving embedding robustness.
- Achieved **90% validation accuracy** on face recognition by generating **128-D embeddings**, validated against baseline models.

Education

University of Waterloo

Sept. 2022 - May 2027

Honours Bachelor's Degree in Mathematics

Waterloo, ON